FEDERAL SPECIFICATION

EVACUATOR, BLADDER AND SYRINGE, BLADDER, EVACUATING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers bladder evacuators suitable for use with resectoscope, cystoscope, and panendoscope sheaths.

1.2 Classication.

1.2.1 *Types and styles.*—Evacuators shall be of the following types and styles, as specified (see 6.1 and 6.4).

Type I.—Ellik.

Type II.—Toomey (syringe).

Style 1.—Glass barrel and tip.

Style 2.—Glass barrel with brass tip.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG–I–526 — Instruments, Dental and Surgical; General Specifications.

QQ–B–626 — Brass, Leaded and Nonleaded; Rods, Shapes, Forgings and Flat Products with Finished Edges (Bars, Flat Wire and Strips).

ZZ–T–831—Tubing; Rubber.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Boxes, Fiber.

PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 601—Rubber; Sampling and Testing.

Fed. Test Method Std. No. 626—Glassware, Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–L–10547 — Liners, Case, Waterproof.

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U.S. Pharmacopoeial Convention Publication:

Pharmacopoeia of the United States.

(Application for copies should be addressed to the Mack Printing Company, Easton, Pa.)

American Chemical Society Publication:

Reagent Chemicals.

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth Street, N.W., Washington 6, D.C.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Glass.* — Glass parts shall be good quality, clear borosilicate glass and shall be capable of withstanding successfully all tests hereinafter specified.

3.1.2 *Metal parts.*—All metal parts shall be fabricated of brass in conformance with composition 22 of Federal Specification QQ–B–626. The brass shall be plated with nickel only in conformance with plating requirements for class 4, composition N of Federal Specification GG–I–526. The plated metal parts shall also comply with the requirements for resistance to boil test of Federal Specification GG–I–526.

3.1.3 *Rubber.*

3.1.3.1 *Bulb.* — Bulb shall be fabricated from natural or synthetic red rubber compound.

3.1.3.2 *Tubing.*—Tubing shall be commercial gum-rubber tubing conforming to the applicable requirements of Federal Specification ZZ–T–831 for grade A tubing.

3.2 Type I.—Type I evacuator shall be of the Ellik design. The evacuator shall consist essentially of a glass receptacle, a rubber bulb, and a short rubber tube with a metal adapter. Style, design, and dimensions of type I evacuator shall be in conformance with figure 1.

3.2.1 *Glass receptacle.*—Receptacle shall be a heavy walled, symmetrically made glass container in the approximate shape of an hourglass (in accordance with figure 1) and shall be suitable for trapping excised tissue removed (evacuated) from the bladder.

3.2.1.1 *Annealing.*—Receptacle shall be examined for strain, as specified in 4.3.7. A moderate amount of evenly distributed longitudinal strain is acceptable. Any ring strain or severe longitudinal strain will not be acceptable.

3.2.1.2 *Alkalinity (solubility).* — Receptacles shall be tested for alkalinity, as specified in 4.3.5. The volume of 0.02 normal (N) sulfuric acid required to neutralize the pink color, which may be observed after the solution has cooled, shall not exceed 1.0 milliliter (ml.).

3.2.1.3 *Resistance to thermal shock.*—When tested as specified in 4.3.6, no breakage shall occur due to temperature changes.

3.2.2 *Rubber bulb.*—Bulb shall be spherical in shape, of 3- or 4-ounce capacity, with smooth inner surface, and shall have a cylindrical neck of heavy reinforced rubber. The neck of the bulb shall fit tightly within the flared end of the glass body connector to insure against leakage during normal use. The bulb shall be of stiff rubber, easily compressible yet sufficiently resilient for the purpose intended. When subjected to the accelerated aging test specified in 4.3.9, the bulb shall show no softening or lack of resilience and shall not become brittle, sticky, or tacky.

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3.2.3 *Rubber tubing.*—Rubber tubing shall be 4½ ± ½ inches in length and shall have an inside diameter of 5/16 ± 1/64 inch and a 3/32 ± 1/64 inch wall thickness.

3.2.4 *Adapter.*—Adapter shall be a metal connector for attachment to cystoscope sheaths and the other end for securing to distal end of rubber tubing as shown in figure 1.

3.3 Type II.—Type II evacuator shall be of the Toomey design, suitable for use with resectoscope, cystoscope, or panendoscope. The evacuator shall consist essentially of a barrel, a plunger, a rubber washer, and an adapter. Design and dimensions of styles 1 and 2 shall be as shown in figures 2 through 4 (see 6.4).

3.3.1 *Barrel.*—Barrel of style 1 evacuators shall be all glass, and the outside surface shall be graduated by marking as shown in figure 2 or, as an alternate, the graduation may be in 5-ml. intervals. Style 2 evacuator barrels shall be glass, ungraduated, with brass cap firmly cemented to the glass. The metal and glass connection shall show no leakage, when tested as specified in 4.3.1, after being subjected to the autoclaving and shock tests (see 4.3.3 and 4.3.4). The inside surface of the barrel shall be finished to a very fine, uniformly ground surface, free from pits, grind marks, or open air lines.

3.3.1.1 *Graduations and other markings.*—The outside surface of the barrel of the style 1 evacuator shall have marked thereon the graduation scale required (see 3.3.1). These markings shall be applied to the barrel in such a manner that they are in line with the flattened side of the flange. Main graduation marks shall be at least 33⅓ percent longer than the subgraduation marks, and the "0" mark of the scale shall register precisely with the extreme distal end of the plunger when pushed to the bottom of the barrel. The serial number shall be applied to the barrel of type II evacuator. All markings shall be clearly legible, and either fused in, fused on, or etched and fused. Evacuators shall be subjected to the permanency of marking test specified in 4.3.8 and shall be considered to pass the test if not more than one half of any graduation line or numeral has been lost, and the area showing complete pigment loss is less than 2 percent of the total pigmented area, and if the legibility of any portion of the markings is not seriously impaired.

3.3.2 *Plunger.*—Plunger shall be hollow. The distal end of the plunger shall be ground flat and at right angles to the axis, and its edges shall be slightly beveled. The beveled edge of style 1 evacuator shall be of permanently dark color, or the plunger may be of dark colored glass throughout its length, or for part of its length from the distal end. The proximal end of the plunger shall extend well out of the barrel as shown in figure 2. The plunger shall show under low power (4 to 8X) magnification a uniform, smooth-ground surface, free from pits, grind marks, or open air lines, and shall so accurately fit its individual barrel as to make the evacuator successfully withstand the fluid tightness test prescribed in 4.3.1 and yet move freely throughout its excursion in the barrel. The plunger shall be permanently marked near the proximal end with a serial number with fused pigment corresponding to that on the barrel.

3.3.3 *Rubber washer.* — Rubber washers shall be fabricated of soft latex rubber and shall be of appropriate size as to fit loosely at distal end of barrel in styles 1 and 2 evacuators as shown in figures 2 and 3.

3.3.4 *Adapter.*—Adapter shall be designed to fit styles 1 and 2 evacuators and shall be constructed as shown in figure 4. The adapter shall successfully withstand the fluid tightness test as specified in 4.3.1. The adapter shall be clearly and legibly marked with the name or registered trademark of the manufacturer.

3.3.5 *Fluid tightness.*—Evacuators shall be subjected to the fluid tightness test as specified in 4.3.1. During the first test there shall not be enough water to pass the plunger or

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the adapter to produce a falling drop. After each successive test, any water that may have passed the plunger or the adapter shall be wiped away and the next test applied, during which there shall not be enough water to pass the plunger or the adapter to produce a falling drop.

3.3.6 *Alkalinity.*—When tested as specified in 4.3.2, the volumes of 0.02N sulfuric acid required to neutralize the pink color, which may be observed after the solution has cooled, shall not exceed 0.16 ml. for style 1 evacuator and 0.09 ml. for style 2 evacuator.

3.3.7 *Resistance to autoclaving.*—Evacuators shall be subjected to the test specified in 4.3.3. At the end of the test the plunger shall be freely movable in the barrel. There shall be no loosening of the cemented cap on the barrel of style 2 evacuators.

3.3.8 *Resistance to thermal shock.*—When tested as specified in 4.3.4, no breakage shall result due to temperature changes. There shall be no loosening of the cemented cap on the barrel of type II evacuators.

3.3.9 *Annealing.*—The barrel and plunger, when examined as specified in 4.3.7, shall not show any ring strain or severe longitudinal strain. A moderate amount of diffused longitudinal strain is acceptable.

3.4 **Identification markings.**—The evacuators shall be clearly and legibly marked with the manufacturer's name or registered trademark.

3.5 **Workmanship.**—Workmanship shall be first class throughout. Evacuators shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**

4.1.1 *For visual examination.*—Sampling procedures for visual examination shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.1.2 *For tests.*

4.1.2.1 *For fluid tightness, permanency of marking, and resistance to autoclaving tests.*—Sample size for each test shall be as indicated for inspection level L3, AQL equals 1.0 percent defective in Appendix to Military Standard MIL–STD–105. The unit of product for inspection purposes shall be one evacuator.

4.1.2.2 *For shock test.*—Sample size for test shall be as indicated for inspection level L3 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero for all sample sizes, and a minimum of five samples shall be taken regardless of lot size. The unit of product for inspection purposes shall be one evacuator.

4.1.2.3 *For alkalinity tests.*—Sample size shall be as specified in the applicable test description (or method).

4.1.2.4 *For bulb accelerated aging and for rubber tube tests.*—Sample size for test shall be as indicated for inspection level L1 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero for all sample sizes, and a minimum of two samples shall be taken regardless of lot size.

4.1.2.5 *For metal parts tests.*—Sample size for each test shall be as indicated for inspection level L2 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero for all sample sizes and a minimum of four samples shall be taken regardless of lot size. The unit of product for inspection purposes shall be one evacuator.

4.2 **Inspection.**—The inspector shall ascertain that the evacuators have been examined and tested in accordance with specification requirements.

4.3 **Tests.**

4.3.1 *Fluid tightness.* — The evacuators shall be assembled with adapter and attached

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in a horizontal position to a water pressure source having a pressure gage. A water pressure of 25 pounds to the square inch shall be used. The water used as a testing fluid shall be let in through the tip of the adapter. The plunger shall be drawn to a point one-fourth of the evacuator capacity where it shall be allowed to remain for 10 seconds. The plunger shall then be drawn back to midway between first position and the full-capacity point of the evacuator where it shall be allowed to remain for a similar length of time. The plunger shall then be drawn back to the point of full capacity of the evacuator where it shall be allowed to remain for a similar length of time.

4.3.2 *Alkalinity.* — Evacuators shall be washed in distilled water. All surfaces shall be brushed to remove loose particles and thoroughly rinsed with the water. Unassembled evacuators, with metal parts removed, shall then be placed in an alkali-free resistant glass beaker; and covered with a solution of freshly boiled distilled water to which has been added 0.2 ml. of United States Pharmacopoeia (U.S.P.) phenolphthalein test solution for each 100 ml. of water. The number of evacuators to be used, the size of beaker, and the volume of water are given in the following table. Evacuators shall be autoclaved at 15 pounds steam pressure for 15 minutes and then removed from the autoclave.

Evacuators	Number of evacuators (not assembled)	Size of beaker	Volume of water
Style 1 (type II) ---	5	4000 ml.	2000 ml.
Style 2 (type II) ---	5	2000 ml.	1475 ml.

4.3.3 *Resistance to autoclaving.*—Evacuators (dry) with the plunger pushed completely in the barrel shall be subjected to autoclave sterilization for 15 minutes at 15 pounds steam pressure.

4.3.4 *Thermal shock test (type II).* — Evacuators shall be transferred suddenly from boiling water into water at 3° to 5° Centigrade (C.). This shock test shall be repeated at least 10 times.

4.3.5 *Alkalinity (solubility) test.* — The solubility shall be determined in accordance with method 320 of Federal Test Method Standard No. 626.

4.3.6 *Thermal shock test.*—The glass receptacles shall be heated in an autoclave at 15 pounds pressure for 15 minutes. Then shall be removed from the autoclave and quickly plunged into water of 3° to 5°C.

4.3.7 *Annealing strain.*—Glass parts shall be examined for strain with a polariscope that is constructed as described in method 420 of Federal Test Method Standard No. 626.

4.3.8 *Permanency of marking test.*—Evacuators shall have metal parts removed and then be autoclaved at 15 pounds per square-inch gage (p.s.i.g.) (250°Fahrenheit(F.)) completely immersed in an open beaker containing hydrochloric acid solution. The hydrochloric acid solution shall consist of reagent in the proportion of 10 ml. of 35-38 percent hydrochloric acid conforming to the requirements of The American Chemical Society in 1 liter of distilled water for two periods of 30 minutes each. Each 30-minute period shall begin at the instant the temperature of the solution in the autoclave reaches 245°F. as measured by a thermocouple immersed in the solution. At the end of each period the evacuators shall be removed and rinsed with cold tap water, then dried by wiping with cellulose tissue. Evacuators shall then be examined for loss of pigmentation.

4.3.9 *Bulb accelerated aging test.*—The accelerated aging test for the rubber bulbs shall be determined in accordance with method 7111 of Federal Test Method Standard No. 601, except that the rubber bulb shall be tested intact, and the tensile strength and ultimate elongation tests will not be performed.

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4.4 Supplier responsibilities for inspection. —The supplier shall perform examinations and tests set forth in this specification to substantiate conformance of supplies to specification requirements. The examinations and tests shall be at the expense of the supplier. Suppliers shall have adequate test facilities or shall arrange for test facilities acceptable to the Government to be used for executing specific prescribed tests. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government upon the Government's request, at any time, or from time to time, for a period of 2 years after the delivery of the supplies to which these records relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing protection used are defined in Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Unit of issue.* — One evacuator, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue, indicated in table I, shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

TABLE I.—*Packaging quantities*

Federal stock number	Unit package	Intermediate package	Exterior container
6515–328–7500 or 6515–328–8100	1 unit	Not required	48 units

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636 or PPP–B–676. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table III.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table I.

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5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Unit packages—Military.*—Each unit package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Exterior container—Military.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except 6.2(c), 6.2(d), and 6.5 shall not be acceptable. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

5.3.3 *Unit packages—Nonmilitary.*—Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.4 *Exterior container—Nonmilitary.*—In addition to any special marking specified in the contract or order, exterior containers shall be marked in accordance with Federal Standard No. 123.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Types and styles required (see 1.2.1).
(*c*) Selection of applicable levels of packaging and packing (sec 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Federal Supply Catalog, Department of Defense Section, for Medical Material:

6515–328–7500 EVACUATOR, BLADDER, ELLIK.

6515–328–8100 SYRINGE, BLADDER EVACUATING, TOOMEY.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 For Military Medical Supply Agency procurement.—Either style 1 or style 2 of type II is acceptable (see 1.2 and 3.3).

6.5 Transportation description.—The transportation description applicable to this item is:

Pumps, Suction or Pressure
With or Without Appliances for Drainage or other Treatment of Body Cavities.

Carload minimum weight 12,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor Volume minimum weight 12,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army — M
Navy — Md
Air Force.

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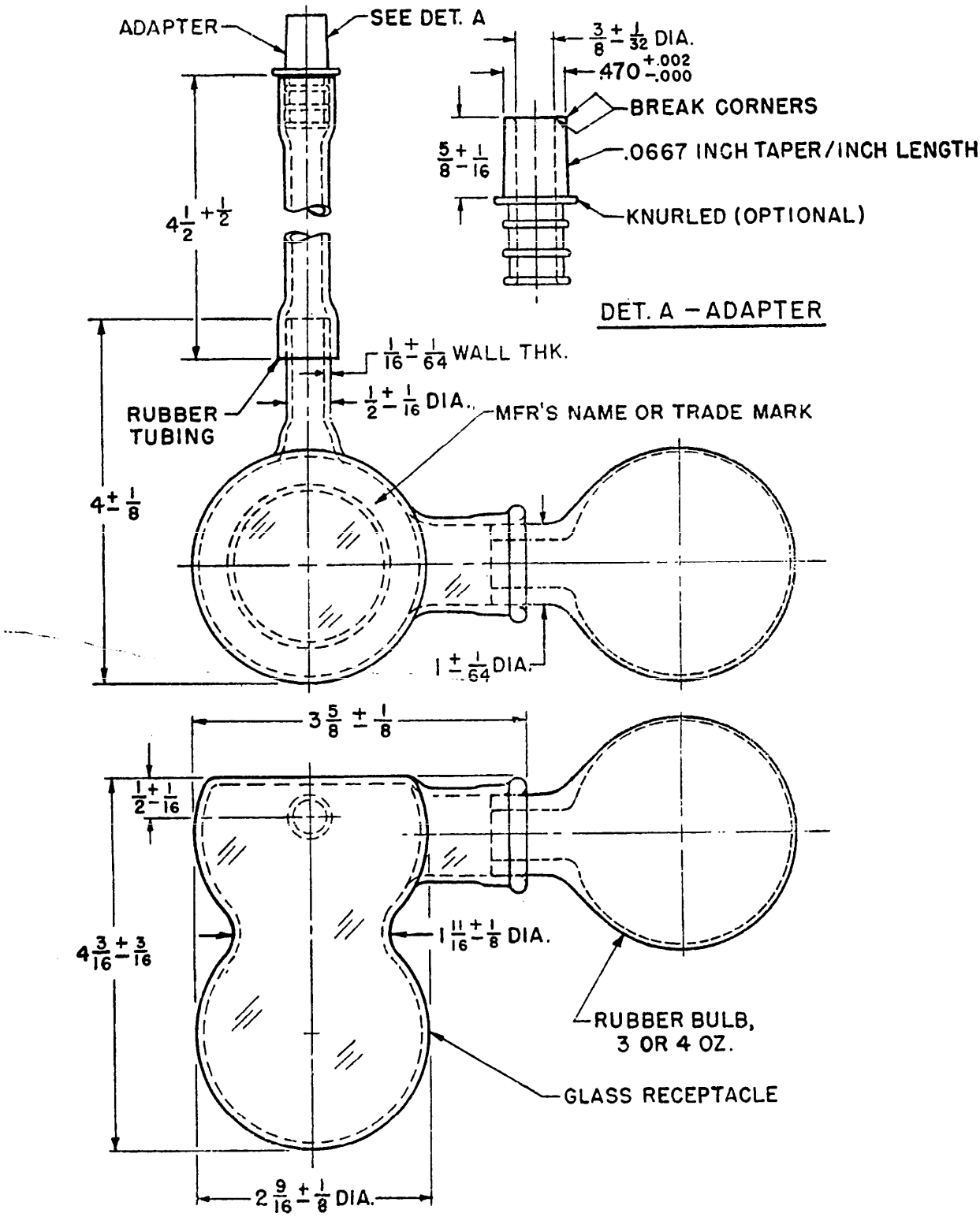


FIGURE 1.—Type I.

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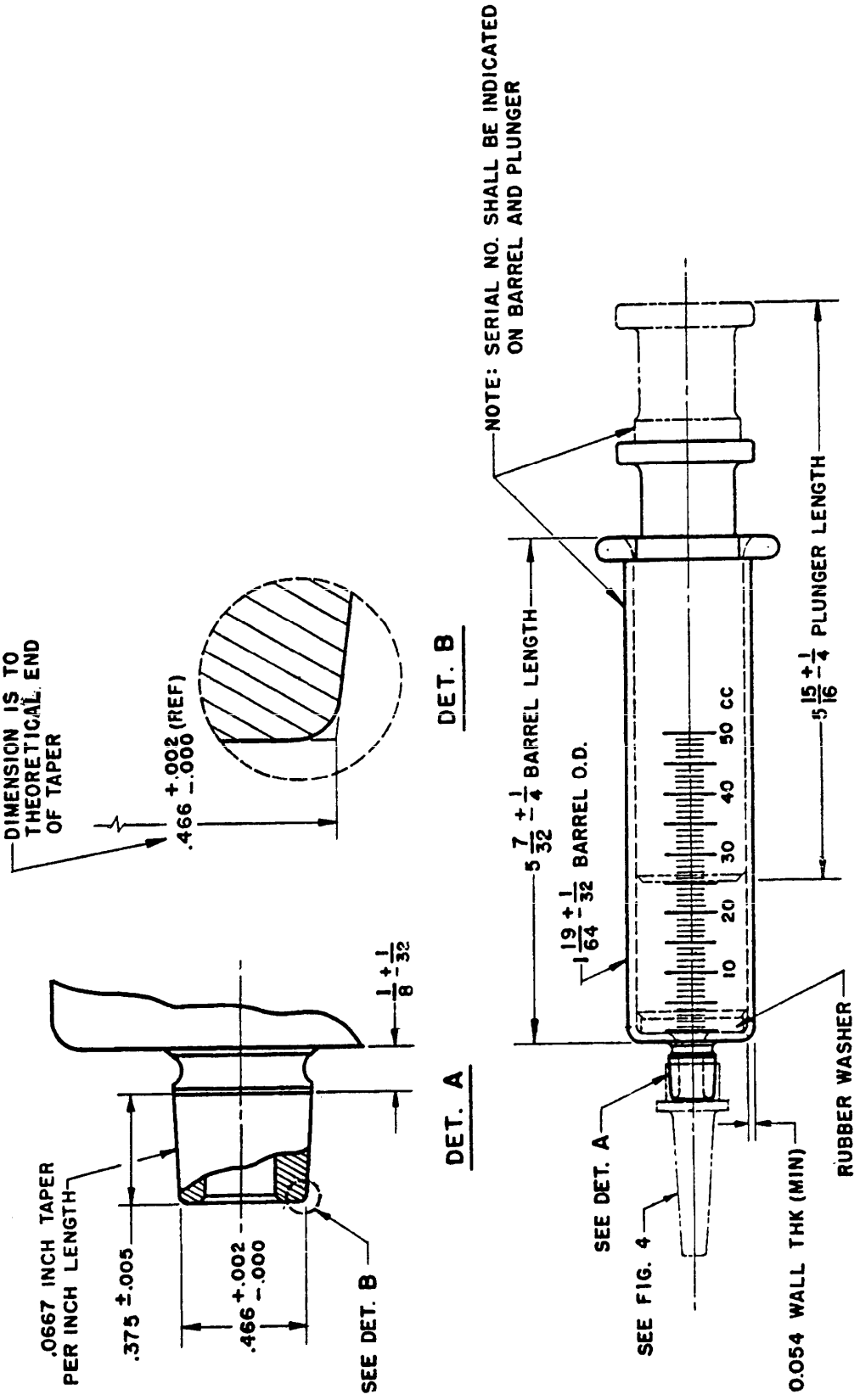


FIGURE 2.—Type II, style 1.

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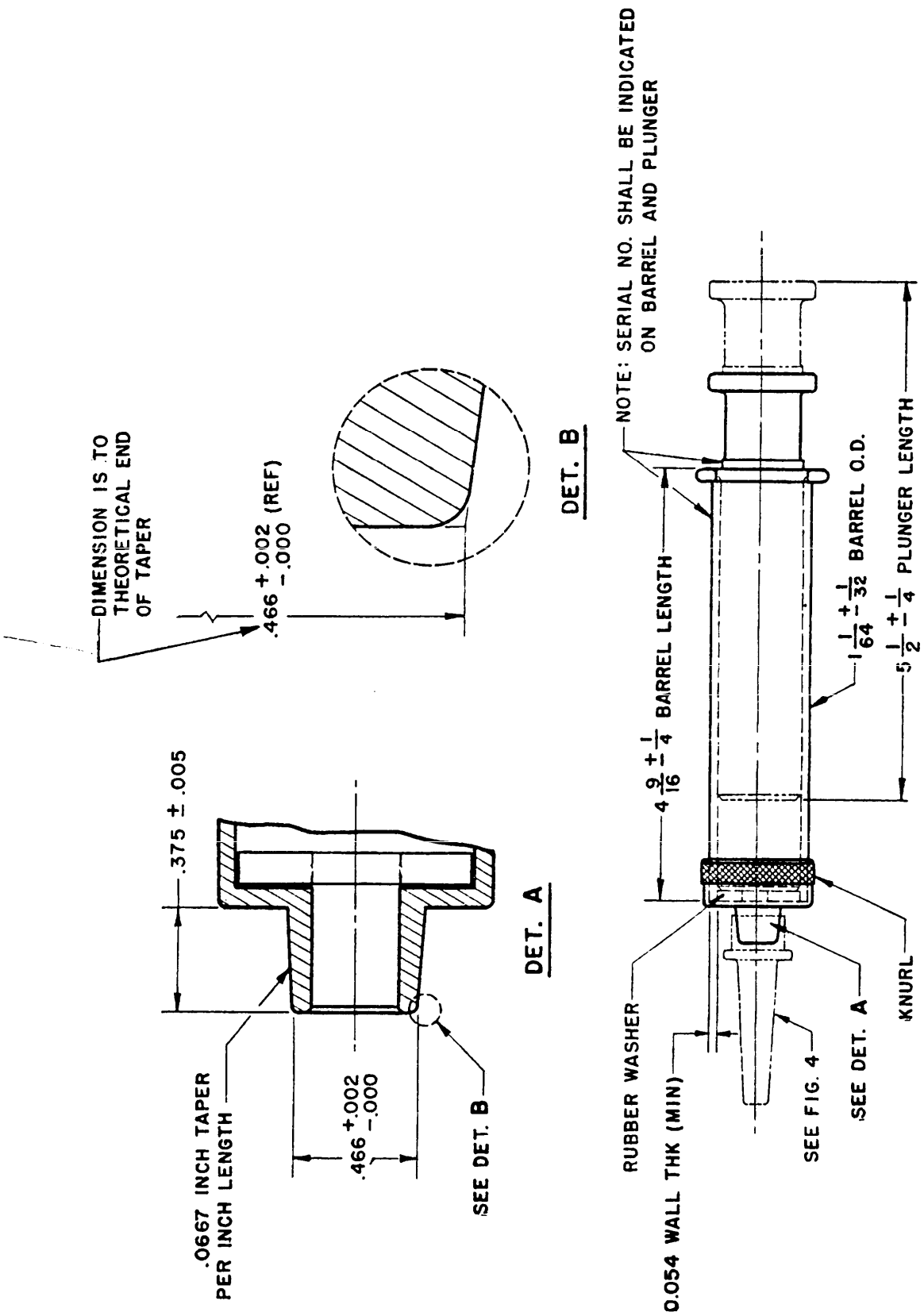


FIGURE 3.—Type II, style 2.

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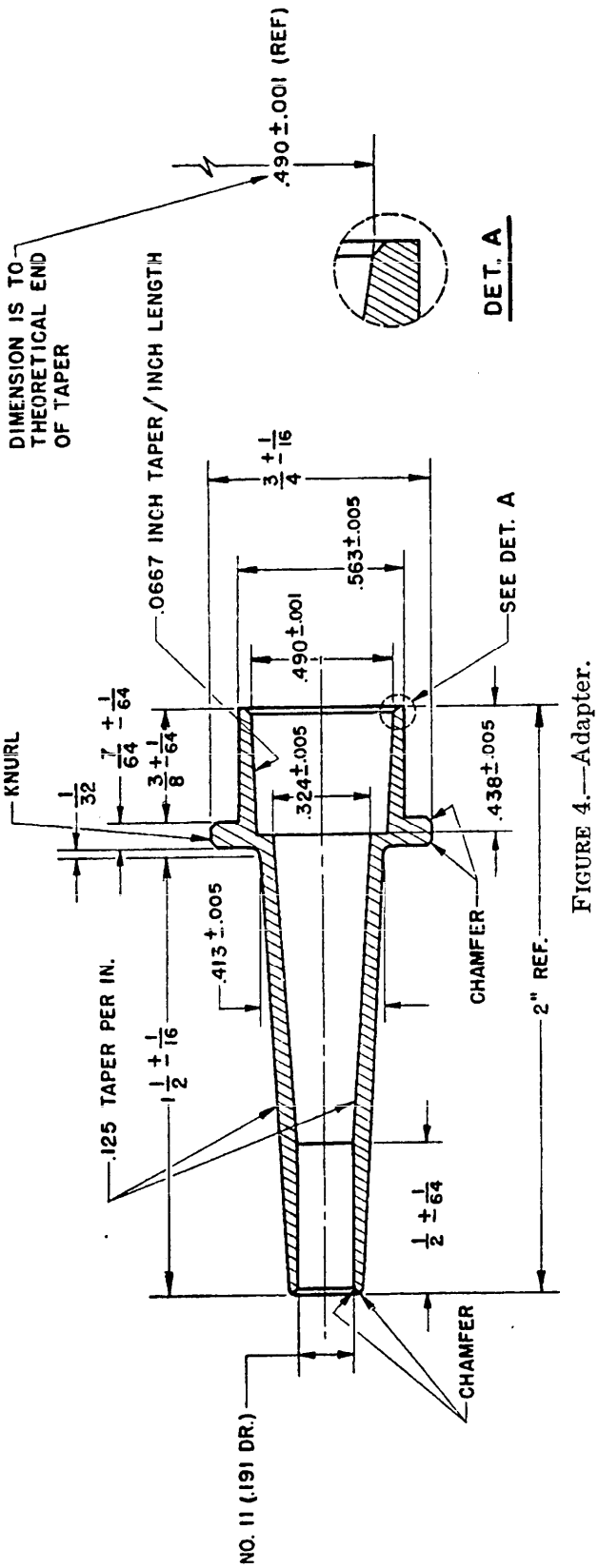


FIGURE 4.—Adapter.

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